



Boiler Strategy Group

Financial Process Transformation



Krenicki Center for Business
Analytics and Machine Learning

Multi-agent AI framework within SAP S/4HANA to automate receivables matching, dispute resolution, and behavioral risk insights. AI-driven agents reduce manual receivable operations while supporting in predictive financial decision.

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ABSTRACT

Manual data entry and reactive dispute management create significant inefficiencies in **traditional Accounts Receivable (AR) operations** for large retail enterprises. This project designs a conceptual, multi-agent AI ecosystem within **SAP S/4HANA Cloud** to modernize the Order-to-Cash workflow.

By integrating three specialized AI models - an **AR Ledger Agent** for autonomous payment matching, a **Dispute Resolution Agent** for automated case validation, and a **Behavioral Insights Agent** for predictive risk scoring - the team streamlined complex financial data flows. The proposed architecture significantly reduces administrative workloads, accelerates dispute resolution, and improves bad-debt reserve accuracy.

Ultimately, this study presents a scalable framework for integrating agentic AI to optimize enterprise finance and **shift human effort** from data searching to **strategic review**.

BUSINESS PROBLEM

Manual AR matching, dispute handling, and risk review **delay cash collection** and **increase operational cost** in large enterprises.

- High-volume payments require manual remittance investigation and invoice matching.
- Disputes are handled reactively, increasing cycle time and operational overhead.
- Risk segmentation for bad debt reserves remains judgment-heavy and inconsistent.

Client Challenges:

- High-Volume Manual Matching
- Reactive Dispute Management
- Subjective Risk Segmentation



ACCOUNTS RECEIVABLE AGENT

The goal of the agent is to match and clear incoming payments, reducing manual investigation effort up to 90%



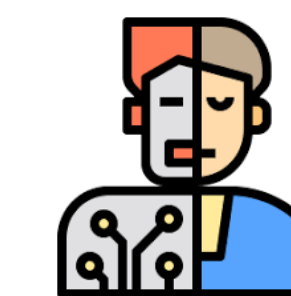
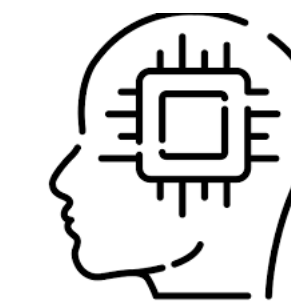
Before flowchart of ACH/Wire/EFT L5 process

Key Outputs: Automated payment matching, ledger postings, exception flags for unmatched payments, and accountant review proposals.

Yes (>90% Confidence): **AUTO-CLEARs** directly to S/4HANA Ledger

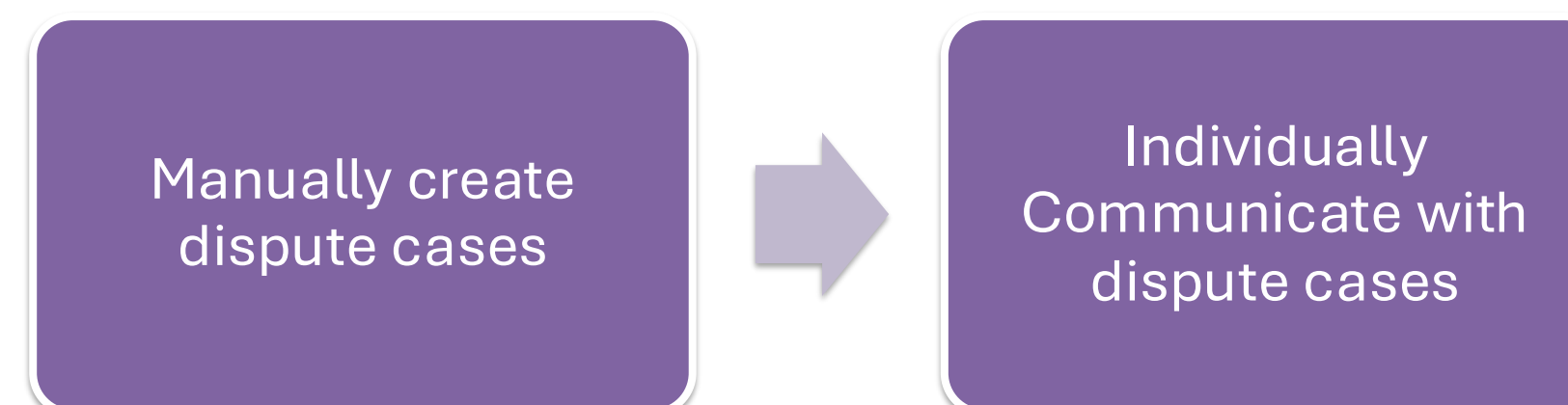
No (Low Confidence): Generates **'Predictive Proposal'** for AR Accountant Review

Exception (Missing Data): **FLAGS** invalid items for strategic investigation



DISPUTE RESOLUTION AGENT

The goal of the agent is to scan invoices for errors automatically along with detecting incorrect charges and suggesting fixes



Before flow of Dispute Case Opening and Management (manual dispute handling process)

Key Outputs: Validated dispute cases, recommended corrective actions (credit memos/adjustments), and prioritized dispute resolution workflow.

Scans invoices and receipts for errors automatically

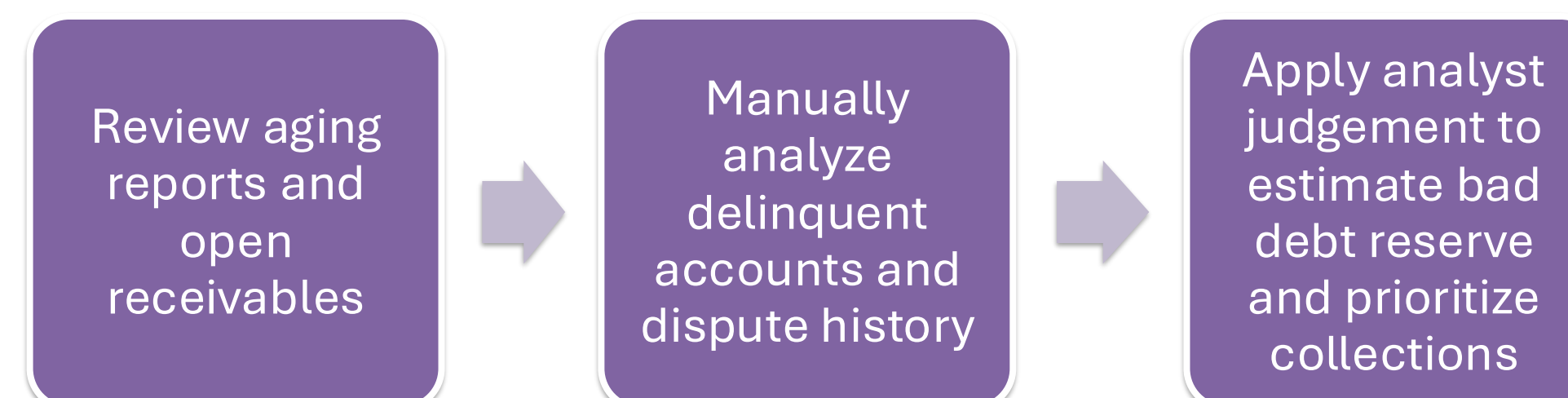
DETECTS incorrect charges and suggests fixes like creation of credit memos

Conducts faster dispute resolution allowing to improve vendor relationships.



BEHAVIORAL INSIGHTS AGENT

The goal of the agent is to analyze historical financial behavior, disputes, and exposure patterns to predict late payments and default risk



Before flow of Bad Debt Reserve and Collections Risk Review (manual risk analysis process).

Key Outputs: Customer risk segmentation, prioritized collections worklist, behavioral risk drivers, and alerts for financial review.

FLAGS likely late payers using payment and dispute patterns.

Prioritizes accounts with **CLEAR** risk drivers for outreach.

RAISES anomalies or data gaps for finance review.



LIMITATIONS

- Time and scope:** Conceptual design only, no live pilot or end-to-end enterprise deployment.
- Scalability:** Requires strong data pipelines, SAP integration, governance, and ongoing monitoring to scale across regions and workflows.
- Other AI + human oversight needed:** Document understanding, explainability, audit controls, and human approvals remain essential for exceptions and high-impact decisions.

CONCLUSION

- AI agents significantly improve financial process efficiency by **automating** repetitive tasks.
- The system shifts finance roles from manual investigation to strategic oversight and **decision making**.
- Despite automation, **human expertise remains essential** for handling exceptions, validating AI recommendations, and making critical financial decisions.

FUTURE WORK & IMPACT

- Methodological Scalability:** Our core impact is a highly adaptable evaluation framework. By extracting ML metrics from SAP documentation to build clear "Before vs. After" flowcharts, we can design integration blueprints for any SAP AI agent.
- Future Expansion:** Applying this repeatable methodology beyond Accounts Receivable will allow enterprises to systematically identify, visually map, and validate automation opportunities.

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